

Basic Queries

Select all columns:

```
SELECT * FROM employees;
```

Select specific columns:

```
SELECT first_name, last_name, salary FROM employees;
```

Filter by condition:

```
SELECT * FROM employees WHERE salary > 8000;
```

Filter by multiple conditions:

```
SELECT * FROM employees WHERE department_id = 50 OR department_id = 80;
```

Filter by pattern matching (LIKE):

```
SELECT * FROM employees WHERE first_name LIKE 'M%';
```

Sort results:

```
SELECT first_name, last_name, salary FROM employees ORDER BY salary DESC;
```

Limit results:

```
SELECT * FROM employees FETCH FIRST 10 ROWS ONLY;
```

String Functions

Concatenate columns:

```
SELECT first_name || ' ' || last_name AS "Full Name" FROM employees;
```

Convert to uppercase/lowercase:

```
SELECT UPPER(first_name), LOWER(last_name) FROM employees;
```

Find string length:

```
SELECT LENGTH(first_name) FROM employees;
```

Find position of a character:

```
SELECT INSTR(last_name, 'a') FROM employees;
```

Extract part of a string:

```
SELECT SUBSTR(first_name, 1, 2) FROM employees;
```

Date Functions

Current date:

```
SELECT SYSDATE FROM dual;
```

Add months:

```
SELECT ADD_MONTHS(SYSDATE, 3) FROM dual;
```

Find months between two dates:

```
SELECT MONTHS_BETWEEN(SYSDATE, hire_date) FROM employees;
```

Get last day of the month:

```
SELECT LAST_DAY(SYSDATE) FROM dual;
```

Get next occurrence of a weekday:

```
SELECT NEXT_DAY(SYSDATE, 'Monday') FROM dual;
```

Round and truncate dates:

```
SELECT ROUND(SYSDATE, 'MONTH'), TRUNC(SYSDATE, 'MONTH') FROM dual;
```

Aggregations

Count rows:

```
SELECT COUNT(*) FROM employees;
```

Average salary:

```
SELECT AVG(salary) FROM employees;
```

Sum of salaries:

```
SELECT SUM(salary) FROM employees;
```

Minimum and maximum salary:

```
SELECT MIN(salary), MAX(salary) FROM employees;
```

Numeric Functions

Power function:

```
SELECT POWER(4, 2) FROM dual; -- Output: 16
```

Square root:

```
SELECT SQRT(121) FROM dual; -- Output: 11
```

Ceiling value:

```
SELECT CEIL(3.5) FROM dual; -- Output: 4
```

Floor value:

```
SELECT FLOOR(-3.2) FROM dual; -- Output: -4
```

Modulo (remainder):

```
SELECT MOD(10, 3) FROM dual; -- Output: 1
```

Round number:

```
SELECT ROUND(745.126, 2) FROM dual; -- Output: 745.13
```

Truncate number:

```
SELECT TRUNC(745.126, 2) FROM dual; -- Output: 745.12
```

Absolute value:

```
SELECT ABS(-2.3) FROM dual; -- Output: 2.3
```

Character Functions

Convert to lowercase:

```
SELECT LOWER('HELLO') FROM dual;  -- Output: hello
```

Convert to uppercase:

```
SELECT UPPER('hello') FROM dual;  -- Output: HELLO
```

Capitalize first letter:

```
SELECT INITCAP('hello world') FROM dual;  -- Output: Hello World
```

Concatenation:

```
SELECT CONCAT('Hello', ' World') FROM dual;  -- Output: Hello World
```

Substring extraction:

```
SELECT SUBSTR('Hello', 1, 3) FROM dual;  -- Output: Hel
```

Find string length:

```
SELECT LENGTH('Hello') FROM dual;  -- Output: 5
```

Find position of character:

```
SELECT INSTR('Hello', 'l') FROM dual;  -- Output: 3
```

Left pad string:

```
SELECT LPAD('Hello', 10, '*') FROM dual;  -- Output: *****Hello
```

Right pad string:

```
SELECT RPAD('Hello', 10, '*') FROM dual;  -- Output: Hello*****
```

Trim characters:

```
SELECT TRIM(LEADING 'H' FROM 'Hello') FROM dual;  -- Output: ello
```

Replace characters:

```
SELECT REPLACE('Hello World', 'World', 'SQL') FROM dual;  -- Output: Hello SQL
```

IN and NOT IN:

Find employees in specific departments:

```
SELECT * FROM employees WHERE department_id IN (50, 80);
```

Find employees with specific job IDs:

```
SELECT * FROM employees WHERE job_id IN ('SA_MAN', 'ST_CLERK');
```

Exclude employees from departments:

```
SELECT * FROM employees WHERE department_id NOT IN (50, 90);
```

Substitution Variables:

Filter by department:

```
SELECT * FROM employees WHERE department_id = :enter_dept_id;
```

Joins

Employees and their department:

```
SELECT first_name, last_name, department_id FROM employees
WHERE department_id IN (SELECT department_id FROM departments);
```

Employees and their managers:

```
SELECT first_name, last_name, manager_id FROM employees
WHERE manager_id IN (SELECT manager_id FROM employees);
```

Salary Updates

Increase salary by 10%:

```
SELECT first_name, last_name, salary, salary + salary * 0.10 FROM
employees;
```

Decrease salary by 200:

```
SELECT first_name, last_name, salary, salary - 200 FROM employees;
```